

Business Intelligence Mini Project Report

Customer Churn Prediction using Data Mining

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1 Introduction

Customer churn is one of the most critical challenges faced by organizations in industries such as telecommunications, banking, and subscription-based services. It refers to the phenomenon where customers discontinue using a company's services and switch to competitors. This directly impacts revenue, growth, and long-term sustainability of the business.

From a business perspective, acquiring new customers involves significantly higher costs compared to retaining existing ones. Therefore, organizations are increasingly focusing on predictive analytics to identify customers who are at risk of leaving. Business Intelligence (BI) plays a crucial role in this process by transforming raw data into actionable insights.

This project focuses on applying data mining techniques to predict customer churn using a real-world dataset. The objective is to analyze customer behavior patterns and build a predictive model that can assist organizations in making proactive retention decisions.

2 Problem Definition

The primary objective of this project is to develop a predictive model that can determine whether a customer is likely to churn based on historical data. The problem can be viewed from both business and technical perspectives.

From the business viewpoint, the goal is to minimize customer attrition, improve customer satisfaction, and enhance profitability. Organizations can use churn predictions to design targeted marketing campaigns, offer personalized incentives, and improve service quality.

From the technical perspective, the problem is formulated as a supervised learning task. The input consists of customer-related attributes such as demographic information, service usage patterns, and billing details. The output is a categorical variable indicating whether the customer will churn (Yes or No).

3 Data Mining Task

The given problem falls under the category of classification in data mining. Classification involves assigning predefined labels to instances based on input features. In this case, the

model predicts whether a customer belongs to the churn or non-churn category.

Several classification algorithms can be applied to solve this problem, including Decision Trees, Logistic Regression, and Random Forest. Among these, Decision Trees are particularly suitable due to their interpretability and ability to generate clear decision rules.

Decision Trees work by recursively splitting the dataset based on feature values, creating a tree-like structure where each internal node represents a decision and each leaf node represents an outcome. This makes the model easy to understand for non-technical stakeholders.

4 Dataset Description

The dataset used in this project is the Telco Customer Churn Dataset, which is publicly available on Kaggle. It contains information about customers of a telecommunications company and includes various attributes related to customer demographics, account details, and service usage.

4.1 Key Attributes

Attribute	Description
customerID	Unique customer identifier
gender	Male or Female
tenure	Duration of customer relationship
MonthlyCharges	Monthly subscription cost
TotalCharges	Total amount paid
Contract	Type of contract
InternetService	Type of internet service
PaymentMethod	Mode of payment
Churn	Target variable (Yes/No)

Table 1: Dataset Attributes

The dataset is well-structured and suitable for classification tasks, making it ideal for churn prediction analysis.

5 System Architecture

The system follows a structured pipeline approach consisting of multiple stages, starting from data collection to final prediction.

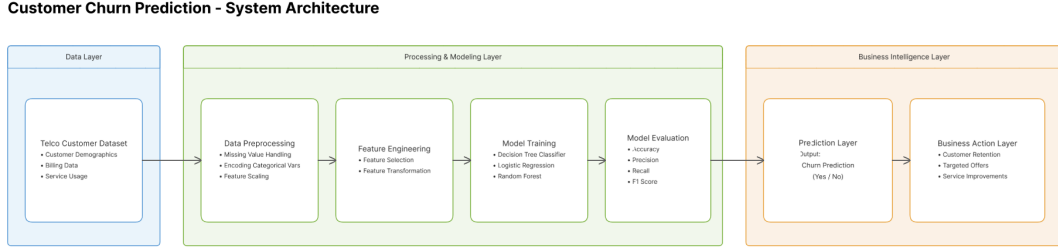


Figure 1: Customer Churn Prediction System Architecture

This architecture ensures a systematic flow of data and helps in building a reliable predictive system. The architecture shown in Figure 1 represents a layered pipeline designed to systematically transform raw customer data into actionable business insights.

The process begins with the **Data Layer**, which consists of the Telco Customer Dataset. This dataset includes essential information such as customer demographics, billing details, and service usage patterns. This raw data serves as the foundation for the entire system.

The next stage is the **Processing and Modeling Layer**, where the core data mining operations are performed. Initially, in the **Data Preprocessing** stage, the dataset is cleaned by handling missing values, encoding categorical variables, and applying feature scaling to ensure consistency. This is followed by **Feature Engineering**, where important features are selected and transformed to improve model performance.

After preprocessing, the refined data is passed to the **Model Training** stage, where machine learning algorithms such as Decision Tree, Logistic Regression, and Random Forest are applied to learn patterns from historical data. The trained model is then evaluated in the **Model Evaluation** stage using performance metrics such as Accuracy, Precision, Recall, and F1-score to ensure its effectiveness.

Finally, the system transitions to the **Business Intelligence Layer**. In the **Prediction Layer**, the trained model predicts whether a customer is likely to churn (Yes/No). Based on these predictions, the **Business Action Layer** enables organizations to take strategic actions such as offering targeted retention schemes, improving services, and enhancing customer engagement.

Overall, this architecture provides a clear and efficient flow from raw data to decision-making, enabling organizations to leverage data-driven insights for reducing customer churn.

6 Data Preprocessing

Data preprocessing is a crucial step in the data mining process. Raw data often contains inconsistencies, missing values, and categorical variables that need to be transformed before model training.

The first step involves handling missing values, either by removing incomplete records or by imputing them using statistical methods such as mean or median values. Next, categorical variables such as gender and payment method are converted into numerical representations using encoding techniques.

Feature scaling is applied to normalize numerical attributes such as monthly charges and total charges. This ensures that all features contribute equally to the model. Finally, the dataset is divided into training and testing subsets to evaluate model performance effectively.

7 Model Building

In this project, a Decision Tree classifier was implemented using the `scikit-learn` library to predict customer churn. The dataset was first preprocessed by handling missing values in the `TotalCharges` column and converting all categorical variables into numerical format using one-hot encoding (`get_dummies`). This ensured that the dataset was suitable for machine learning algorithms.

The dataset was then divided into training and testing sets using an 80:20 split. The training set was used to train the Decision Tree model, while the testing set was used to evaluate its performance on unseen data.

The Decision Tree algorithm works by recursively splitting the dataset based on feature values. At each step, it selects the feature that best separates the data using criteria such as Gini index. This process continues until the stopping condition is met, resulting in a tree-like structure where each path represents a decision rule.

In this implementation, the model was trained using default parameters with a fixed random state to ensure reproducibility. Once trained, the model was used to make predictions on the test dataset.

The choice of Decision Tree is justified due to its interpretability and ability to handle both numerical and categorical data effectively. It also provides a clear understanding of how different features influence the prediction, which is useful for business decision-making.

```

1 # Step 1: Import Libraries
2 import pandas as pd
3 import numpy as np
4 from sklearn.model_selection import train_test_split
5 from sklearn.tree import DecisionTreeClassifier
6 from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
7
8 # Step 2: Load Dataset
9 df = pd.read_csv("demo.csv")
10
11 # Step 3: Basic Cleaning
12 df.drop("customerID", axis=1, inplace=True)
13
14 # Convert TotalCharges to numeric
15 df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors='coerce')
16
17 # Fill missing values
18 df["TotalCharges"] = df["TotalCharges"].fillna(df["TotalCharges"].median())
19
20 # Step 4: Convert categorical to numeric (CORRECT WAY)
21 df = pd.get_dummies(df, drop_first=True)
22
23 # Step 5: Split Features & Target
24 X = df.drop("Churn_Yes", axis=1)
25 y = df["Churn_Yes"]
26
27 # Step 6: Train-Test Split
28 X_train, X_test, y_train, y_test = train_test_split(
29     X, y, test_size=0.2, random_state=42
30 )
31
32 # Step 7: Train Model
33 model = DecisionTreeClassifier()
34 model.fit(X_train, y_train)
35
36 # Step 8: Predict
37 y_pred = model.predict(X_test)
38
39 # Step 9: Evaluation
40 print("Accuracy:", accuracy_score(y_test, y_pred))
41 print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
42 print("\nClassification Report:\n", classification_report(y_test, y_pred))

```

Figure 2: Model Implementation

8 Evaluation Metrics

The performance of the model was evaluated using multiple classification metrics, including accuracy, precision, recall, and F1-score. These metrics provide a detailed understanding of how well the model performs, especially in handling class imbalance.

The model achieved an overall accuracy of approximately 71%, indicating that it correctly classifies a majority of the instances. However, accuracy alone is not sufficient, especially in imbalanced datasets such as churn prediction, where non-churn customers are more frequent than churn customers.

Precision measures how many of the predicted churn cases are actually correct. In this case, the precision for churn prediction is around 45%, which indicates that the model has moderate reliability when predicting churn.

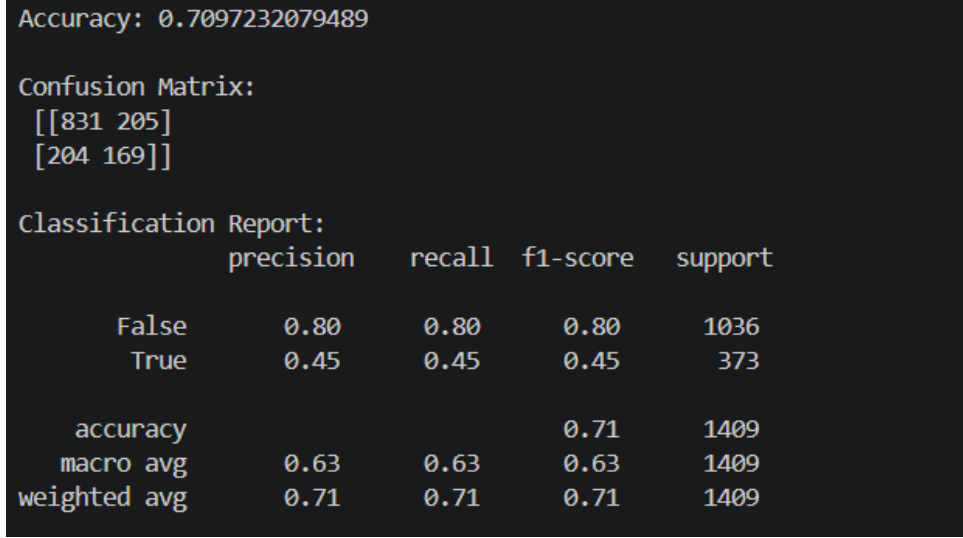
Recall measures how many actual churn cases are correctly identified. The recall value for churn is also around 45%, suggesting that the model is able to detect only a portion of the customers who are likely to leave.

The F1-score provides a balance between precision and recall and is particularly useful in evaluating models on imbalanced datasets. The relatively lower F1-score for churn

indicates that there is scope for improvement in identifying high-risk customers.

Additionally, the confusion matrix provides a clear breakdown of predictions, showing correct and incorrect classifications for both churn and non-churn classes. The model performs significantly better in identifying non-churn customers compared to churn customers, which is a common issue in classification problems with imbalanced data.

Overall, these metrics indicate that while the model performs reasonably well, further improvements can be made using advanced algorithms or data balancing techniques.



```
Accuracy: 0.7097232079489

Confusion Matrix:
[[831 205]
 [204 169]]

Classification Report:
              precision    recall  f1-score   support

      False       0.80      0.80      0.80     1036
      True        0.45      0.45      0.45      373

 accuracy              0.71      1409
 macro avg           0.63      0.63      0.63     1409
weighted avg           0.71      0.71      0.71     1409
```

Figure 3: Model Output

9 Results and Discussion

The Decision Tree model achieved an overall accuracy of approximately 71%, indicating that it is able to correctly classify a majority of customer churn cases. The confusion matrix shows that out of 1409 test instances, 831 non-churn customers were correctly identified, while 169 churn customers were correctly predicted. However, the model also produced some misclassifications, with 205 non-churn customers incorrectly predicted as churn and 204 churn customers incorrectly predicted as non-churn.

The classification report further highlights the model's performance. The precision and recall for non-churn customers are both around 80%, indicating that the model is highly reliable in identifying customers who are likely to stay. In contrast, the precision and recall for churn customers are approximately 45%, suggesting that the model struggles to accurately detect all potential churn cases.

This imbalance in performance is primarily due to the dataset being skewed towards non-churn customers, which causes the model to favor the majority class. As a result,

while the model performs well overall, its ability to identify churn customers is relatively limited.

From a business perspective, the model still provides valuable insights. It effectively captures patterns in customer behavior, such as the impact of monthly charges, contract type, and tenure on churn probability. Customers with higher monthly charges are more likely to churn, whereas those with long-term contracts and longer tenure tend to remain loyal.

These insights can be leveraged by organizations to design targeted retention strategies, such as offering incentives to high-risk customers or improving service quality for specific segments. Although the model can be further improved using advanced techniques, it serves as a strong baseline for understanding and predicting customer churn.

10 Conclusion

Customer churn prediction is an important application of Business Intelligence and data mining. By leveraging historical data and predictive modeling techniques, organizations can make informed decisions and reduce customer attrition.

This project demonstrates how classification algorithms can be effectively used to solve real-world business problems and enhance operational efficiency.

11 Future Enhancements

Future work can focus on improving model accuracy by using advanced machine learning techniques such as ensemble models and deep learning. Integration with real-time systems and visualization dashboards can further enhance usability.

12 References

- Kaggle - Telco Customer Churn Dataset
- Han, J. & Kamber, M. Data Mining Concepts and Techniques